

ELEC5305 Project Proposal

1. Project Title

Comparative Evaluation of Spectral Subtraction and Wiener Filtering for Speech Enhancement under Different Noise Conditions

2. Student Information

Full Name: Jinyuan Liu

Student ID (SID): 550284873

GitHub Username: jliu0709

GitHub Project Link: <https://github.com/jliu0709/elec5305-project-550284873>

GitHub Pages Link: <https://jliu0709.github.io/elec5305-project-550284873/>

3. Project Overview

Background noise can reduce speech clarity and affect everyday communication and audio recordings. This project aims to implement and compare two speech enhancement methods, spectral subtraction and Wiener filtering, using MATLAB. Clean speech selected from Mini LibriSpeech will be mixed with MATLAB-generated white noise and office noise from the DEMAND database at different input signal-to-noise ratios (SNRs). Both methods will process the noisy signals in the short-time Fourier transform (STFT) domain before reconstructing the enhanced audio. Performance will be evaluated through SNR improvement and processing time, supported by spectrograms and informal listening. The central research question is how noise type and input SNR affect the SNR improvement and processing time of spectral subtraction and Wiener filtering.

4. Background and Motivation

Speech enhancement aims to suppress unwanted noise while preserving useful speech information. Spectral subtraction and Wiener filtering are established approaches to this problem. Spectral subtraction removes an estimated noise spectrum from the noisy speech spectrum, but it can introduce artificial tonal sounds commonly described as “musical noise.” Wiener filtering applies frequency-dependent gains based on estimated speech and noise power. Both approaches depend on noise estimation, and inaccurate estimates can result in residual noise or speech distortion.

These limitations motivate investigating their performance under different noise conditions.

I chose this topic because it connects course concepts, including framing, windowing, frequency-domain processing and signal reconstruction, with a practical audio application. Implementing these methods in MATLAB provides a manageable project without requiring neural-network training or specialised hardware. Using a fixed selection of public audio recordings will also make the experiments reproducible. Rather than proposing a new algorithm, the project will provide a controlled comparison and explain the strengths and limitations of each method within the tested conditions.

5. Proposed Methodology

The project will use a custom MATLAB Live Script supported by functions for data preparation, speech enhancement and evaluation. Twenty clean speech clips covering several speakers will be selected from the Mini LibriSpeech development subset. White noise will be generated in MATLAB, and office noise will be obtained from the DEMAND database. One fixed recording channel will be used for the office noise. All signals will be processed as mono audio at 16 kHz. Noise will be scaled to produce input SNRs of 0, 5 and 10 dB, giving six noise conditions for each speech clip. File selections, noise segments and random seeds will be documented.

The noisy signals will be divided into overlapping frames, multiplied by a Hann window and transformed using the STFT. An initial noise-only segment will provide an estimate of the noise power spectrum. Spectral subtraction will subtract this estimate from the noisy speech power spectrum, applying a non-negative spectral floor. Wiener filtering will calculate gains using speech power estimated from the noisy observations and the estimated noise power. Both methods will retain the noisy phase and reconstruct the enhanced signals using inverse transformation and overlap-add.

Both algorithms will use identical mixtures, STFT settings and initial noise estimates. Parameters will be fixed before the final comparison. Clean speech will be used only to construct mixtures and evaluate results, not to calculate enhancement gains. SNR improvement will be measured over the aligned speech portion, excluding the initial noise-only segment. Processing time will be measured on the same computer, excluding file loading, plotting and playback. Results will be averaged across clips for each condition. Spectrograms and informal listening will support qualitative discussion of residual noise and distortion. The small speech sample and reliance on

initial noise estimates will be acknowledged as limitations.

6. Expected Outcomes

The project is expected to deliver working MATLAB implementations of spectral subtraction and Wiener filtering, together with a reproducible evaluation workflow. Tables and plots will compare SNR improvement and processing time across the selected noise conditions. Audio examples and spectrograms will illustrate changes before and after enhancement. The analysis will compare the two methods in terms of SNR improvement and computational cost, without assuming that either method will perform better in every condition. Spectrograms and informal listening will provide supplementary observations rather than a formal assessment of perceived speech quality or intelligibility. The public GitHub repository will include the Live Script, supporting functions, parameter settings, selected-file lists and a README with reproduction instructions. Dataset sources will be credited, and audio redistribution will follow the relevant licences. A GitHub Pages website will present the written report and a short demonstration video.

7. Timeline (Weeks 1–13)

Week	Task
------	------

1–2	Consider Project Topic
-----	------------------------

3–5	Literature review and dataset collection
-----	--

6–9	Initial implementation and testing
-----	------------------------------------

10–11	Optimisation and evaluation
-------	-----------------------------

12–13	Final report and GitHub documentation
-------	---------------------------------------

8. References

[1] M. Gupta, R. K. Singh, and S. Singh, “Analysis of optimized spectral subtraction method for single channel speech enhancement,” *Wireless Personal Communications*, vol. 128, no. 3, pp. 2203–2215, Feb. 2023, doi: [10.1007/s11277-022-10039-y](https://doi.org/10.1007/s11277-022-10039-y).

[2] C. Plapous, C. Marro, and P. Scalart, “Improved signal-to-noise ratio estimation for speech enhancement,” *IEEE Transactions on Audio, Speech, and Language*

Processing, vol. 14, no. 6, pp. 2098–2110, Nov. 2006, doi:

[10.1109/TASL.2006.872621](https://doi.org/10.1109/TASL.2006.872621).

[3] R. K. Jaiswal, S. R. Yeduri, and L. R. Cenkeramaddi, “Single-channel speech enhancement using implicit Wiener filter for high-quality speech communication,” *International Journal of Speech Technology*, vol. 25, no. 3, pp. 745–758, Sep. 2022, doi: [10.1007/s10772-022-09987-4](https://doi.org/10.1007/s10772-022-09987-4).

[4] V. Panayotov, G. Chen, D. Povey, and S. Khudanpur, “Librispeech: An ASR corpus based on public domain audio books,” in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, South Brisbane, Australia, 2015, pp. 5206–5210, doi: [10.1109/ICASSP.2015.7178964](https://doi.org/10.1109/ICASSP.2015.7178964).

[5] J. Thiemann, N. Ito, and E. Vincent, “The Diverse Environments Multi-channel Acoustic Noise Database (DEMAND): A database of multichannel environmental noise recordings,” *Proceedings of Meetings on Acoustics*, vol. 19, no. 1, Art. no. 035081, 2013, doi: [10.1121/1.4799597](https://doi.org/10.1121/1.4799597).